

RAG Chatbot

Setup & Testing Guide

1. Prerequisites

Before running this project, ensure the following are installed on your machine:

Tool	Version	Download
Docker Desktop	Latest	https://www.docker.com/products/docker-desktop
Git (optional)	Any	https://git-scm.com
Gemini API Key	—	https://aistudio.google.com

No Python installation is required on your machine — everything runs inside Docker.

2. Setup Steps

Step 1 — Get the project files

Extract the provided rag-chatbot.zip to a folder on your machine, e.g. C:\Projects\rag-chatbot or ~/projects/rag-chatbot.

Step 2 — Add your Gemini API key

Open the .env file in the project root with any text editor and replace the placeholder:

```
GEMINI_API_KEY=your_actual_key_here
```

Get a free key from: <https://aistudio.google.com> → Get API Key

Step 3 — Build the Docker image

Open a terminal (PowerShell on Windows, Terminal on Mac/Linux), navigate to the project folder and run:

```
docker build -t rag-chatbot .
```

This will take 3–5 minutes the first time (downloads Python, PyTorch, and all dependencies). Subsequent builds use cache and take seconds.

Step 4 — Run the container

Windows (PowerShell):

```
docker run -p 8000:8000 --env-file .env -v ${PWD}/data:/app/data rag-chatbot
```

Mac / Linux:

```
docker run -p 8000:8000 --env-file .env -v $(pwd)/data:/app/data rag-chatbot
```

The server is ready when you see: Uvicorn running on http://0.0.0.0:8000

Step 5 — Open the app

Open your browser and go to: <http://localhost:8000>

3. Using the Chatbot

1. Click "Choose File" and select a plain .txt document from your computer
 2. Click "Upload Document" — wait for the success message
 3. Type a question about your document in the input box and press Send
 4. The bot answers based on your document. Ask follow-up questions freely.
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